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## Survival and Growth of Juvenile Snake River Fall Chinook Salmon Exposed to Constant and Fluctuating Temperatures

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**Abstract.**—The incipient lethal temperature (ILT) and critical thermal maximum (CTM) methods are used to set temperature limits for fish. However, the standard application of these methods does not always match the temperature regimes that fish experience in the wild. We used alternative methods to determine the thermal tolerance thresholds of juvenile fall-run Chinook salmon *Oncorhynchus tshawytscha* exposed to the temperature regimes that are common in the entrainment pools that form along the shoreline of the Snake River when flows are altered to meet electric power demand. A modified CTM test with a steady temperature rise (1.5°C/h) showed that one-half the fish died when temperatures reached 27.4–27.9°C and that survival at 25°C was highly variable; the average time to the first death was 9.1 h, varying from 1.7 to 22.5 h. Over 30 d, 99.8% of the fish in the constant temperature regimes (14–22°C) survived. In the fluctuating temperature regimes (which varied from 10–14°C to 22–27.5°C), overall survival was 97.3%; however, only 83.0% and 88.9% survived in the groups that reached daily maximums of 27°C and 27.5°C, respectively. Growth over 30 d in the constant thermal regimes was nearly twice as high as that in the fluctuating regimes, even when daily average temperatures were similar. The maximum growth was 1.9%/d in terms of fork length [FL] and 11.2%/d in terms of weight (WT) at a constant 20°C. The lowest growth occurred in the two groups exposed to daily temperatures of 27°C or more, namely, 0.7–0.8% (FL) and 2.7–3.4% (WT). The results of this study suggest that thermal tolerance tests that expose juvenile fall Chinook salmon to thermal regimes that match the field conditions in entrainment pools along the shoreline of the Snake River provide higher temperature criteria than the standard ILT and CTM methods.

Fish are frequently entrapped in pools off main stream channels when water levels fluctuate because of anthropogenic activities (Bain et al. 1998; Jones and Stuart 2008). For example, water levels can fluctuate rapidly downstream of a hydroelectric dam as water flow is ramped up and down in response to electric power needs (Hunter 1992), an operation referred to as load following. Fish and other aquatic organisms entrapped from these rapid water level fluctuations can be isolated from the main stream channel for periods of time ranging from hours to days (McMichael et al. 2005, 2006; Jones and Stuart 2008). Shallow entrainment pools are susceptible to solar warming because they are isolated without a replenishing flow; temperatures become elevated as substrate and surface water absorb solar energy. Fish caught in entrainment pools when flows fluctuate are exposed to increasing temperature regimes they would otherwise not experience in the main stream channel.

Originally developed to investigate thermal effects

on fish physiology (Fry 1947; Brett 1956), the incipient lethal temperature (ILT) and the critical thermal maximum (CTM) methods are commonly used by management agencies to set lethal temperature levels for fish in the wild (Brungs and Jones 1977). The ILT method involves acclimating fish to a range of temperatures followed by the rapid transfer and exposure to a range of temperatures. Mortalities are noted at each exposure temperature (e.g., Brett 1952), and the ILT is the temperature that results in 50% mortality of the test animals at a given acclimation temperature. Because the ILT incorporates exposure time to a range of temperatures, it is a good indicator of thermal tolerance and is commonly used to set temperature standards within state water quality criteria (Becker and Genoway 1979). However, the ILT method may not extrapolate well to the natural environment because it is rare for fish in natural habitats to experience abrupt changes in water temperature. Further, the ILT tests are of relatively short duration (usually  $\leq 7$  d), which may not be long enough to examine physiological changes that reflect the long-term effects of temperature exposure.

The CTM method involves exposing fish to steady

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and rapid temperature increase (e.g.,  $18^{\circ}\text{C}/\text{h}$ ) until each fish loses equilibrium or dies (Coutant 1973; Becker and Genoway 1979). The CTM method is advantageous over the ILT method because it measures acute temperature tolerance and can be completed quickly using simple equipment and with fewer fish (Selong et al. 2001). However, the CTM method does not incorporate exposure time into the thermal tolerance limits because temperatures during the procedure are constantly changing. Further, as with the ILT method, extrapolation of the CTM results to the natural environment may be problematic because the rate of temperature change used in most CTM methods is much faster than temperature regimes experienced by fish in the natural environment.

SNAKE RIVER fall-run Chinook salmon *Oncorhynchus tshawytscha* that are listed under the U.S. Endangered Species Act (NMFS 1992) spawn downstream of Hells Canyon Dam, a large load-following dam on the Snake River (Groves and Chandler 1999). Their fry emerge and rear along the shorelines from early spring to early summer and then begin seaward movement as subyearlings (Connor et al. 2002). The rearing period of juvenile Snake River fall Chinook salmon makes them susceptible to entrapment in pools that form when releases of water from the Hells Canyon Dam fluctuate in response to load-following operations. Water in entrapment pools of the Snake River below Hells Canyon Dam experience slow warming ( $1\text{--}2^{\circ}\text{C}/\text{h}$ ) to  $3\text{--}9^{\circ}\text{C}$  above the flowing water temperature of the river ( $10\text{--}14^{\circ}\text{C}$ , depending on time of year), followed by decreases in temperature ( $3^{\circ}\text{C}/\text{h}$ ) as the entrapment pool is reconnected to the river (Figure 1).

We conducted a laboratory study in 2008 to evaluate juvenile fall Chinook salmon survival and growth under temperature regimes that were relatively consistent with those observed in entrapment pools along the Snake River, as depicted in Figure 1. Using empirical data, we exposed fish to a heating rate and temperature profile more realistic of conditions fish experience in the Snake River entrapment pools, in contrast to standard ILT or CTM temperature rates. Short-term thermal tolerance tests were conducted using a modified CTM method, followed by additional long-term (30 d) thermal exposure tests involving both constant and fluctuating thermal regimes representative of the Snake River entrapment pools. Both survival and growth were monitored because growth integrates many physiological processes and behavioral changes in fish (NAS 1972). The three objectives of this study were to (1) determine the upper temperature at which juvenile fall Chinook salmon lose equilibrium or die when exposed to an increase in temperature that is consistent with daily temperatures found in entrapment

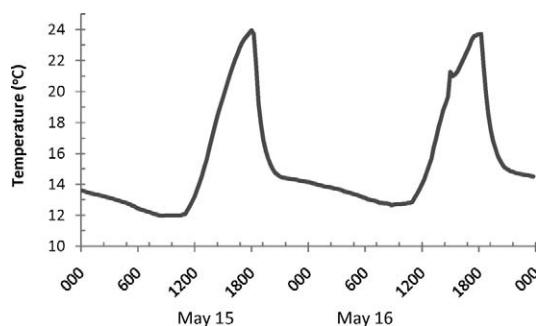


FIGURE 1.—Water temperature profile (measured in 15-min increments) of an entrapment pool during a 2-d period in May 2007 at Durham Bar in the Snake River (350.2 kilometers upstream of its confluence with the Columbia River). During this period, the mean rate of temperature increase (1200 to 1900 hours) was about  $1.5^{\circ}\text{C}/\text{h}$ , and the mean rate of temperature decrease (1900 to 2400 hours) was about  $2.9^{\circ}\text{C}/\text{h}$ .

pools of the Snake River, (2) determine the optimum temperature for growth and survival of juvenile fall Chinook salmon held for 30 d at a series of constant temperatures that varied from  $14^{\circ}\text{C}$  to  $22^{\circ}\text{C}$ , and (3) determine growth and survival of juvenile fall Chinook salmon that are exposed to temperatures that fluctuate daily from minimum temperatures of  $10\text{--}14^{\circ}\text{C}$  to maximum temperatures of  $22\text{--}27.5^{\circ}\text{C}$ . The results of this study will be of interest to resource managers faced with understanding the impact of elevated temperature on juvenile salmonids found in environments that are not represented by standard thermal tolerance tests.

## Methods

### Source and Care of Fish

In April 2008, about 2,500 subyearling fall Chinook salmon were captured (beach seines) in the Hells Canyon Reach of the Snake River at rearing areas associated with Pine Bar (river kilometer [rkm] 366.3, as measured from its confluence with the Columbia River) and Russell Bar (rkm 353.7). Fish were transported to the Pacific Northwest National Laboratory in Richland, Washington, where they were sorted by size. The smaller fish were retained for the study because subyearling fall Chinook salmon ( $<60$  mm fork length) appear to be the most susceptible to entrapment (unpublished data of the authors). Water temperature of the transport tank (472.5 L) was about  $8.5^{\circ}\text{C}$  during the 5-h trip to the laboratory. When the fish arrived at the laboratory, they were placed in an 830-L outdoor circular holding tank (about  $7.2^{\circ}\text{C}$ ); flow into the circular tank was about 56 L/min. A groundwater well supplied the water for all phases of this study. After fish had been in the circular tank for 24 h, the temperature was adjusted to  $10^{\circ}\text{C}$  at a rate of

2°C/d. Within 24 h of arriving at the laboratory, fish were feeding to satiation on frozen brine shrimp and pellet fish food (Bio Diet Grower, Bio-Oregon, Inc), and within 4 or 5 d they fed exclusively on pellets (1.0–1.5 mm) twice per day; this same pellet size was used throughout all phases of this study.

#### *Thermal Tolerance Trials*

*Steady temperature rise.*—After about 1 week, about 180 fish were moved from the circular tank and distributed in groups of 60 into three indoor troughs (63 × 30 cm by 15 cm deep; 30.2 L). At the time the fish were transferred, the temperature of the troughs was 9.3°C, and the temperature of the circular tank was 9.5°C. Once fish were in the troughs, we adjusted temperatures 2°C/d until temperatures of the troughs one through three were 10°C, 12°C, and 14°C, respectively. Fish were held at these temperatures for at least 14 d before being used in the thermal tolerance trials (Becker and Genoway 1979; Smith and Fausch 1997; Selson et al. 2001). During the acclimation period, fish were fed to satiation twice per day; feeding was suspended at least 24 h before initiating thermal tolerance trials.

The maximum thermal tolerance trials with steady temperature rise were conducted in three 37-L glass aquaria (hereafter test chambers), with each test chamber set at a starting temperature of 10°C, 12°C, or 14°C. Water was delivered to each test chamber from an elevated head tank (113 L) at about 7.2 L/min; at this flow rate the test chamber turned over about once every 4 min. Water level in the test chamber was maintained with a screened overflow. Temperature of each head tank was controlled by a computer program (written in CRBasic and implemented via LoggerNet; Campbell Scientific Inc, Logan, Utah) that operated two solenoid valves (Irritrol Utra Flow 700–1; Irritrol Systems, Riverside, California) that controlled delivery of chilled water (down to 7.2°C) or heated water (up to 30°C) as needed to maintain target temperatures; actual water temperatures were normally within 0.2°C of target temperatures.

The night before each trial, one group of 10 fish from each trough was weighed (wet weight to nearest 1 mg; WT) and measured (fork length to the nearest 1 mm; FL), and transferred to the test chamber that was at the same temperature (10, 12, or 14°C). The thermal tolerance trial was initiated the following morning by uploading a computer program to the data logger. The computer program automatically and continuously raised the water temperature within each test chamber 1.5°C/h through the trial; the average rate of temperature increase of Snake River entrapment pools in 2007 was about 0.6–0.8°C/h and the maximum rate was

1.9°C/h (thermograph data collected by the authors), suggesting 1.5°C/h would be representative of the most rapid thermal scenarios fish would encounter in the Snake River entrapment pools. Observations and recordings of the time and temperature when fish mortality or loss of equilibrium occurred were made hourly until temperatures reached 19°C, after which observations were made at least once every 15 min (usually more frequently) until all the fish in the test chamber died, at which time the trial was terminated. Loss of equilibrium was defined as when the fish was still alive but upside down and not able to right itself. A continuous recording video system (Open Eye HDDRX) connected to monochrome video cameras (one per chamber, Ikegami ICD-4224) was also used to document behavior of test fish. Each maximum thermal tolerance trial was repeated five times at each of the three acclimation temperatures.

*Temperature rise and hold.*—We also tested survival and monitored behavior of fall Chinook salmon acclimated to three temperatures (10, 12, or 14°C) and held at a sustained temperature of 25°C after a gradual temperature rise; 25°C was about the upper lethal level reported in the literature (Brett 1952; Orsi 1971). In these trials, about 10 fish were moved from acclimation tanks to each of the three test chambers the night before the trial was conducted. The sustained exposure trial was initiated the next morning by uploading a computer program to the data logger that continuously increased the water temperature 1.5°C/h in each chamber until it reached 25°C. Temperatures of the test chambers were maintained at 25°C for at least 70 h (up to 74 h), at which time the number of fish alive was recorded. Each trial was repeated three times at each of the three acclimation temperatures.

*Data analysis.*—The design of the maximum thermal tolerance experiment using a steady temperature rise had five blocks (trials) and three treatments (10, 12, and 14°C acclimation temperatures). Within each trial and acclimation temperature we determined the average temperature associated with four endpoints: first fish to lose equilibrium, first dead, 50% dead, and 100% dead (hereafter endpoint temperatures). The endpoint temperatures were compared using two-way analysis of variance (ANOVA), the main effects being trial and acclimation temperature. The ANOVA interaction effect between trials and acclimation temperature could not be assessed because there was only one observation associated with each of the four endpoints per trial and acclimation temperature. Orthogonal linear contrasts of the mean endpoint temperatures associated with acclimation temperature, after removing the block effect, were used to evaluate the linear slope and the lack of fit to the simple linear

model, and quadratic contrasts were used to evaluate the lack of fit to the simple linear model. When the relationship between the endpoint temperature and acclimation temperature was nonmonotonic (as indicated by a significant quadratic contrast), a Tukey's honestly significantly different (HSD) multiple comparison test was used to compare all possible pairs of endpoint temperatures and acclimation temperatures. All statistical tests were done using statistical software (JMP, version 5.1, SAS Institute, Inc.), and  $\alpha = 0.05$ .

Total percent survival within the sustained thermal tolerance experiment was calculated as the number of fish that remained alive at the termination of the experiment divided by the number of initial test fish. The number of hours to the first death was confirmed either through reviewing the video recordings or from direct observations. No formal statistical tests were conducted on these data. Rather, the total percent survival and number of hours to first death from each of the three trials were qualitatively compared among the three acclimation temperatures.

#### *Constant and Fluctuating Water Temperatures*

*Survival and growth.*—Subyearling fall Chinook salmon were exposed to a series of thermal regimes (both constant and fluctuating) using 20 exposure systems operated simultaneously. Each exposure system consisted of one mixing tank (113 L), four 22-L buckets (hereafter, test containers) where fish were held, two solenoid valves (to control hot and cold water input to the mixing tank), four acrylic flowmeters (Model 7511, King Instrument Company, Garden Grove, California;  $\pm 6\%$  accuracy) to control water flow to the test containers, four temperature sensors (one in each test container; Model 107-L, Campbell Scientific Inc.;  $\pm 0.2^\circ\text{C}$  accuracy over the range of  $0$ – $50^\circ\text{C}$ ), and one data logger (model CR1000) and controller (Model SDM-CD16AC, Campbell Scientific) housed within a weather-resistant enclosure. The solenoid valves were powered by 110-V AC power through 24-V AC transformers. The data acquisition and control system (including data loggers, controllers, and temperature sensors) operated on 12-V DC from sealed rechargeable batteries (model PS100, Campbell Scientific Inc.). Heated (up to  $30^\circ\text{C}$ ) and chilled (down to  $7.2^\circ\text{C}$ ), oxygenated well water flowed by gravity into the mixing tanks, where a low volume pump mixed water. Flow out of the mixing tanks to the test containers and then out of the test containers to a central drain was by gravity. Flow rate to each test container was manually regulated daily by adjusting the flowmeter to about 1.5 L/min; this resulted in a turnover rate of once every 15 min. Each test container was covered with netting to prevent fish from jumping

TABLE 1.—Thermal regimes to which juvenile fall Chinook salmon were exposed.

| Temperature regime ( $^\circ\text{C}$ ) | Mean daily temperature ( $^\circ\text{C}$ ) | Acclimation (d) |
|-----------------------------------------|---------------------------------------------|-----------------|
| Constant                                |                                             |                 |
| 14                                      | 14.1                                        | 17              |
| 15                                      | 15.2                                        | 16              |
| 16                                      | 16.1                                        | 16              |
| 17                                      | 17.1                                        | 15              |
| 18                                      | 18.0                                        | 15              |
| 19                                      | 19.0                                        | 14              |
| 20                                      | 20.0                                        | 14              |
| 21                                      | 21.0                                        | 13              |
| 22                                      | 22.0                                        | 13              |
| Fluctuating                             |                                             |                 |
| 10–22                                   | 12.8                                        | 19              |
| 10–23.5                                 | 13.6                                        | 19              |
| 10–25                                   | 14.6                                        | 19              |
| 12–22.5                                 | 14.0                                        | 18              |
| 12–24                                   | 14.7                                        | 18              |
| 12–25.5                                 | 15.4                                        | 18              |
| 12–27                                   | 16.4                                        | 18              |
| 14–23                                   | 15.4                                        | 17              |
| 14–24.5                                 | 16.0                                        | 17              |
| 14–26                                   | 16.7                                        | 17              |
| 14–27.5                                 | 17.4                                        | 17              |

out and a screened overflow to prevent fish from escaping through the outlet. Dissolved oxygen was at least 90% air-saturation during the entire study; this was verified periodically using a dissolved oxygen meter (YSI Model 70, Yellow Springs, Colorado). All 20 exposure systems were located outside within a covered area that resulted in fish being exposed to natural diurnal light cycles typical of their rearing period. Opaque screens (black landscape cloth) between a set of two exposure systems minimized fish disturbance during feeding and cleaning of test containers. As much as possible, activities around the exposure systems, including care and feeding of fish in the test containers, were done through slits cut in the screens.

The 20 exposure systems were divided into 9 constant thermal regimes and 11 fluctuating thermal regimes (Table 1). The constant thermal regimes ( $C_X$ , where  $X$  = temperature) varied from  $14^\circ\text{C}$  to  $22^\circ\text{C}$  in  $1^\circ\text{C}$  increments. The daily fluctuating thermal regimes ( $F_{Y-Z}$ , where  $Y$  = the base temperature of 10, 12, or  $14^\circ\text{C}$  and  $Z$  the maximum daily temperature of  $22$ – $27.5^\circ\text{C}$ , depending on the regime) increased at a rate of  $1.5^\circ\text{C/h}$  to the maximum temperature and then immediately began to decrease at a rate of  $3^\circ\text{C/h}$  back to the base temperature. These maximum temperatures were selected to represent worst-case temperature scenarios that subyearling fall Chinook salmon might experience in the entrapment pools in the Snake River. Both constant and fluctuating thermal regimes were computer-controlled using the data logger and control-

ler. Temperature sensors recorded water temperature in each test container and the average of all four test containers was used to control water input to the mixing tanks. Temperature data were recorded to the data logger every 4 s, and data loggers were downloaded about once daily. Manual adjustments to flow rates or temperatures of input water were made periodically to minimize variance around the target temperatures; hourly difference between target and test temperatures averaged  $0.13^{\circ}\text{C}$  and varied from  $-0.15$  to  $+0.41^{\circ}\text{C}$  for all 20 exposure systems over the 30-d study.

About 2 weeks after arriving at the laboratory, about 2,000 fish were moved from the circular holding tank to the 80 test containers within the 20 exposure systems. Each test container received about 25 fish (range, 22–26); fish density in the test containers at the beginning of the experiment varied from about 1.6 to 2.1 g/L and at the end of the experiment varied from about 3.3 to 8.6 g/L. On the day the fish were moved, the temperature of the holding tank and all exposure systems was  $10^{\circ}\text{C}$ . Acclimation of fish within each exposure system was begun immediately by adjusting temperatures  $2^{\circ}\text{C}/\text{d}$  (for those with a base above  $10^{\circ}\text{C}$ ) until the starting temperatures were achieved. Starting temperatures for the constant exposure regimes were the same as the test temperatures (i.e.,  $C_{14}$  was acclimated to  $14^{\circ}\text{C}$ ,  $C_{15}$  to  $15^{\circ}\text{C}$ , etc.). Starting temperatures for fish in the fluctuating regimes were the same as their base temperatures (i.e., 10, 12, or  $14^{\circ}\text{C}$ ). Fish were acclimated to their respective starting temperature for 13–19 d, depending on the treatment, before the experiment was initiated (i.e., upon initiation, temperatures for the constant group remained unchanged, whereas temperatures for the fluctuating group were adjusted up and down; Table 1). During the acclimation period, fish were fed twice daily. Uneaten food and fecal matter was siphoned out of the test containers once per day. Observations of fish behavior, including feeding activity, and mortalities were made daily.

The initial weight and fork length for each fish was obtained on May 8 and 9, 2008. This was accomplished by anesthetizing fish in the test container using tricaine methanesulfonate at 0.05 g/L of water, quickly weighing and measuring each fish, placing them in a recovery container, and returning fish to the test containers after recovery. Total time from when fish were immobilized to when fish were returned to their test container varied from 5 to 13 min.

Just past midnight on May 10, 2008, the thermal exposures were initiated by uploading system-specific computer programs to each exposure system. Exposure to the constant and fluctuating thermal regimes

continued for 30 d when the experiment was terminated. Fish care during the 30-d experiment was similar to care during the acclimation period (i.e., daily cleaning of test containers) except that fish were fed to satiation and observed at least 5 times per day (0700, 1000, 1300, 1600, and 1800 hours). Feeding levels were adjusted daily to account for growth of an “average” fish, which resulted in fish receiving about 8% of their wet body weight per day; uneaten food in each test container at the end of the day indicated fish were feeding to satiation. Final fork lengths and weights of each fish were obtained on the day the experiment was terminated.

**Data analysis.**—Unadjusted survival was calculated as the difference between the initial numbers of fish placed in each test container compared with the final number. This survival value represented all sources of mortality, including fish jumping out of the test containers (either recorded or not recorded), fish injured or killed in the process of caring for the fish (i.e., contacted by siphon when cleaning test containers, dropped during measuring, etc.), and fish found dead in the test container, which were deaths not attributable to injury and assumed to have resulted from the thermal exposures. Therefore, the adjusted survival rate of each test container is based only on mortality due to thermal exposure (i.e., excluding deaths attributable to nonthermal causes).

Average absolute increases in fork length ( $FL_{\text{final}} - FL_{\text{initial}}$ ) and weight ( $WT_{\text{final}} - WT_{\text{initial}}$ ) were calculated for the groups of fish in each of the 80 test containers (Busacker et al. 1990). These two measures of growth were also expressed as an average percent daily increase from initial fork length and weight (i.e., relative growth rate; Busacker et al. 1990). Before statistical analysis, growth data were transformed to the arcsine square root of the percent change. Because of a lack of homogeneity of within class variances even after data transformation, a nonparametric Kruskal–Wallis  $H$ -test was done to compare the average percent daily increase in fork length and weight among the nine constant exposure systems, and a Tukey’s HSD multiple comparison test was done on the 11 fluctuating regimes. In addition, we also used polynomial regression analyses of the transformed data to determine the treatment that produced the optimum percent daily increase in fork length and weight in both constant and fluctuating regimes.

## Results

### *Thermal Tolerance*

**Steady temperature rise.**—The average fork length of the 146 Chinook salmon used in the five upper thermal tolerance trials was 52.1 mm (range, 39.0 to

65.0 mm), 90% of the fish being less than 59.0 mm (95% confidence interval [CI], 51.3–52.3 mm). The average weight was 1.930 g (0.710 to 3.210 g), 90% of the fish being less than 2.600 g (95% CI, 1.836–2.017 g). The fork length and weight did not differ among trials ( $P > 0.05$  for both).

There was no difference among the five trials in the endpoint temperatures associated with a loss of equilibrium ( $F_{4,8} = 2.69$ ,  $P = 0.109$ ), first dead ( $F_{4,8} = 2.89$ ,  $P = 0.094$ ), 50% dead ( $F_{4,8} = 1.87$ ,  $P = 0.209$ ), or 100% dead ( $F_{4,8} = 1.93$ ,  $P = 0.198$ ). Endpoint temperatures associated with a loss of equilibrium and for first dead fit the simple linear model as a function of acclimation temperature (quadratic contrast among acclimation temperatures:  $df = 8$ ,  $P > 0.37$ ). The endpoint temperatures for loss of equilibrium increased significantly with acclimation temperature (linear contrast among acclimation temperatures:  $t = 4.65$ ,  $df = 8$ ,  $P = 0.002$ ; Figure 2). Loss of equilibrium occurred at mean endpoint temperatures of 26.8°C for fish acclimated at 10°C, 27.0°C at 12°C, and 27.3°C at 14°C. The endpoint temperatures for first dead also increased significantly with acclimation temperature (linear contrast among acclimation temperatures:  $t = 5.20$ ,  $df = 8$ ,  $P = 0.001$ ; Figure 2). The mean endpoint temperatures for first dead were 26.9°C for fish acclimated to 10°C, 27.1°C at 12°C, and 27.5°C at 14°C.

There was a nonmonotonic relationship between endpoint temperatures for 50% and 100% dead as a function of acclimation temperature (quadratic contrast among acclimation temperatures:  $t = 3.67$ ,  $df = 8$ ,  $P < 0.014$ ; Figure 2). Half the fish died at mean endpoint temperatures of 27.5°C in the groups acclimated at 10°C, 27.4°C at 12°C, and 27.9°C at 14°C. There was no significant difference in endpoint temperatures between the 10°C and 12°C acclimation groups (Tukey's HSD test:  $df = 8$ ,  $P = 0.735$ ). The endpoint temperature in the 14°C acclimation group was significantly higher than the 10°C (Tukey's HSD test:  $df = 8$ ,  $P = 0.003$ ) and 12°C ( $P = 0.001$ ) acclimation groups. All the fish died at mean endpoint temperatures of 27.8°C for the groups acclimated to 10°C, 27.6°C at 12°C, and 28.0°C at 14°C. There was no significant difference in endpoint temperatures between the 10°C group and the 12°C (Tukey's HSD test:  $df = 8$ ,  $P = 0.363$ ) and 14°C ( $P = 0.080$ ) groups. The endpoint temperature in the 14°C acclimation group was significantly higher than the 12°C acclimation group (Tukey's HSD test:  $df = 8$ ,  $P = 0.010$ ).

**Temperature rise and hold.**—The average fork length for the three trials where Chinook salmon were held for extended periods at 25°C was 63.9 mm ( $N = 93$ , 95% CI = 62.0–64.0 mm); weight averaged 3.393 g

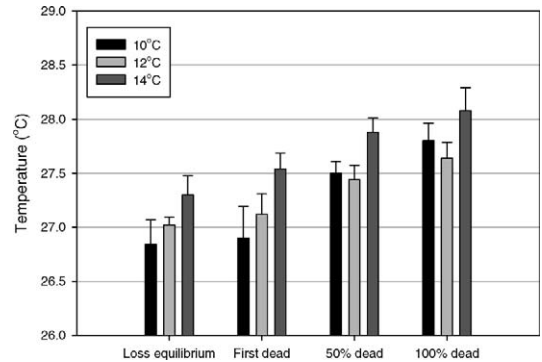


FIGURE 2.—Mean temperatures corresponding to a loss of equilibrium in juvenile fall Chinook salmon, along with the first deaths, 50% mortality, and 100% mortality at three acclimation temperatures (10, 12, and 14°C). The whiskers represent the upper portions of the 95% confidence intervals of the mean of five trials.

( $N = 82$ , 95% CI = 3.069–3.717 g). Both fork lengths ( $F_{2,4} = 52.1$ ,  $P = 0.0014$ ) and weights ( $F_{2,4} = 25.6$ ,  $P = 0.0053$ ) differed among the trials. These fish were substantially larger than the fish used in the previous thermal tolerance trials.

Survival in these trials was highly variable and was not obviously different among acclimation temperatures. The average survival percentage across all three trials was 43% (range, 0–80%) for the group acclimated at 10°C, 70% (50–89%) for those at 12°C, and 74% (60–92%) for those at 14°C. Time to first death averaged 9.1 h (range, 1.7–22.5 h).

#### *Survival and Growth at Constant and Fluctuating Temperatures*

**Constant temperatures.**—The total loss of Chinook salmon over the 30-d experiment from all sources of mortality within the constant thermal regimes was 13 fish (unadjusted survival of 98.5%), but only 2 fish were lost due to thermal exposure (adjusted survival of 99.8%). Not included in this loss were 22 fish that escaped through an unscreened overflow on one test container in the constant 14°C regime the night after the fish were weighed and measured. This container was not included in the analysis.

The maximum growth rate was 1.9% FL/d and 11.2% WT/d at  $C_{20}$  (Table 2). The minimum growth rates occurred at  $C_{14}$  (1.5% FL/d and 7.3% WT/d) and at  $C_{22}$  (1.4% FL/d and 7.3% WT/d). Because of individual variability, the statistical differences overlapped among the different thermal regimes (Table 2). However, based on a Kruskal–Wallis test, the growth rate within  $C_{20}$  was statistically larger than the growth rate for  $C_{14}$  (FL:  $P = 0.031$ ; WT:  $P = 0.011$ ),  $C_{15}$  (FL:

TABLE 2.—Mean (SD) initial and final fork lengths (FL) and weights for juvenile fall Chinook salmon exposed for 30 d to constant thermal regimes varying from 14°C to 22°C. Daily percent increases with different letters are significantly different (Kruskal–Wallis nonparametric test,  $\alpha = 0.05$ ).

| Constant nominal temperature (°C) | FL (mm)     |             | FL gain (%/d) | Weight (g)    |               | Weight gain (%/d) |
|-----------------------------------|-------------|-------------|---------------|---------------|---------------|-------------------|
|                                   | Initial     | Final       |               | Initial       | Final         |                   |
| 14                                | 50.9 (1.08) | 74.5 (1.51) | 1.5 yx        | 1.468 (0.100) | 4.666 (0.275) | 7.3 yx            |
| 15                                | 52.4 (1.13) | 77.0 (1.77) | 1.6 yx        | 1.538 (0.079) | 5.399 (0.381) | 8.4 yx            |
| 16                                | 53.7 (0.63) | 79.7 (1.00) | 1.6 zyx       | 1.644 (0.032) | 6.015 (0.219) | 8.9 zyx           |
| 17                                | 54.2 (0.31) | 82.8 (1.13) | 1.8 zy        | 1.709 (0.043) | 6.817 (0.266) | 10.0 zyx          |
| 18                                | 54.6 (1.51) | 82.9 (2.42) | 1.7 zyx       | 1.746 (0.118) | 7.032 (0.476) | 10.1 zyx          |
| 19                                | 55.2 (0.83) | 84.7 (1.81) | 1.8 zy        | 1.835 (0.089) | 7.589 (0.463) | 10.5 zy           |
| 20                                | 53.3 (0.95) | 83.8 (1.64) | 1.9 z         | 1.656 (0.118) | 7.226 (0.497) | 11.2 z            |
| 21                                | 54.5 (1.58) | 82.2 (2.43) | 1.7 zyx       | 1.822 (0.165) | 6.879 (0.575) | 9.3 zyx           |
| 22                                | 53.1 (1.26) | 74.9 (0.78) | 1.4 x         | 1.649 (0.143) | 5.256 (0.155) | 7.3 x             |

$P = 0.038$ ; WT:  $P = 0.048$ ), and  $C_{22}$  (FL:  $P = 0.001$ ; WT:  $P = 0.004$ ).

Regression analysis of the daily percent increase in fork length and weight showed that a spline model could be fit to both data sets (Figure 3). The maximum daily increase in fork length occurred at 20.4°C (95% CI = 20.0–20.7°C; Table 3) and took the form of

$$Y = \begin{cases} 0.68 + 0.06X & \text{if } X \leq 20.38^{\circ}\text{C} \\ 1.89 - 0.33(X - 20.38) & \text{if } X > 20.38^{\circ}\text{C}, \end{cases}$$

(1)

where  $X$  is the average temperature.

The temperature associated with the maximum daily growth rate in weight occurred at 19.9°C (95% CI = 19.7–20.2°C; Table 3), which was not significantly different from the temperature interval associated with the greatest growth in fork length ( $P = 0.6976$ ). The model for the daily growth rate of weight was

$$Y = \begin{cases} -1.40 + 0.64X & \text{if } X \leq 19.94^{\circ}\text{C} \\ 11.33 - 1.96(X - 19.94) & \text{if } X > 19.94^{\circ}\text{C}. \end{cases}$$

(2)

*Fluctuating temperatures.*—The total loss of fish over the 30-d experiment within the fluctuating regimes due to all sources of mortality was 45 Chinook

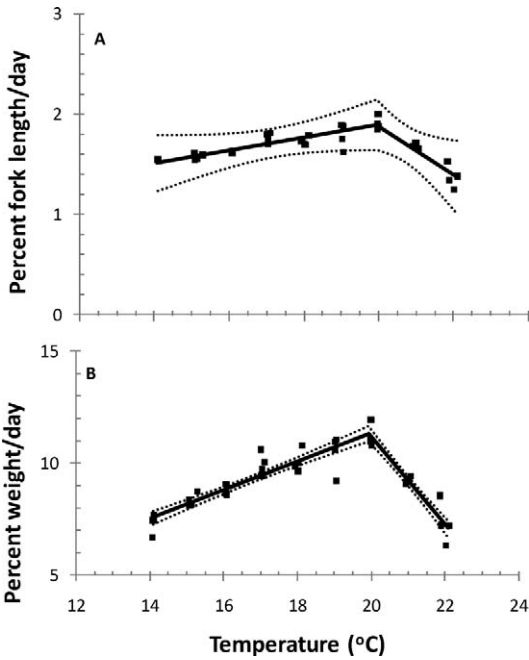


FIGURE 3.—Daily percent increases in (A) the fork length and (B) the weight of juvenile fall Chinook salmon exposed to a range of constant temperatures for 30 d. The squares represent the observed values, the solid lines the expected values based on regression analysis, and the dotted lines the corresponding 95% confidence intervals.

TABLE 3.—Regression results based on the spline model of the daily percent change in the fork length (FL) and weight of juvenile fall Chinook salmon as a function of the average temperature for constant temperature regimes of 14–22°C.

| Parameter or statistic          | FL gain (%/d)  | Weight gain (%/d) |
|---------------------------------|----------------|-------------------|
| <b>Best-fit values</b>          |                |                   |
| Intercept ( $\beta_0$ )         | 0.68           | −1.4              |
| Slope1 ( $\beta_1$ )            | 0.06           | 0.64              |
| X0 (join point)                 | 20.4           | 19.9              |
| Slope2 ( $\beta_2$ )            | −0.33          | −2.0              |
| <b>95% confidence intervals</b> |                |                   |
| Intercept ( $\beta_0$ )         | 0.44 to 0.92   | −3.8 to 0.95      |
| Slope1 ( $\beta_1$ )            | 0.05 to 0.07   | 0.50 to 0.78      |
| X0 (join point)                 | 20.0 to 20.7   | 19.7 to 20.2      |
| Slope2 ( $\beta_2$ )            | −0.43 to −0.23 | −2.3 to −1.6      |
| <b>Goodness of fit</b>          |                |                   |
| df                              | 31             | 31                |
| $R^2$                           | 0.841          | 0.861             |
| Mean square error               | 0.0007         | 0.005             |
| Total number of values          | 35             | 35                |

TABLE 4.—Mean (SD) initial and final fork lengths (FL) and weights for juvenile fall Chinook salmon exposed for 30 d to fluctuating thermal regimes. Daily percent increases with different letters are significantly different (Kruskal–Wallis nonparametric test,  $\alpha = 0.05$ ).

| Fluctuating<br>nominal<br>temperature (°C) | FL (mm)     |             | FL gain<br>(%/d) | Weight (g)    |               | WT<br>gain (%/d) |
|--------------------------------------------|-------------|-------------|------------------|---------------|---------------|------------------|
|                                            | Initial     | Final       |                  | Initial       | Final         |                  |
| 10–22                                      | 51.6 (1.25) | 70.2 (1.60) | 1.2 zyx          | 1.445 (0.129) | 3.853 (0.307) | 5.6 zyx          |
| 10–23.5                                    | 51.8 (1.35) | 70.0 (1.09) | 1.2 zyx          | 1.486 (0.106) | 3.834 (0.268) | 5.3 zyx          |
| 10–25                                      | 51.2 (1.35) | 67.4 (1.32) | 1.1 zyx          | 1.379 (0.131) | 3.396 (0.207) | 4.9 zyx          |
| 12–22.5                                    | 51.3 (1.49) | 71.7 (1.55) | 1.3 zyx          | 1.422 (0.096) | 4.187 (0.371) | 6.5 zy           |
| 12–24                                      | 52.3 (0.74) | 72.0 (1.52) | 1.3 zyx          | 1.521 (0.074) | 4.309 (0.163) | 6.1 zyx          |
| 12–25.5                                    | 52.1 (0.83) | 69.6 (1.48) | 1.1 zy           | 1.454 (0.039) | 3.771 (0.199) | 5.3 zyx          |
| 12–27                                      | 51.3 (0.84) | 64.1 (0.98) | 0.8 yx           | 1.428 (0.093) | 2.873 (0.206) | 3.4 yx           |
| 14–23                                      | 53.1 (1.18) | 76.2 (1.50) | 1.4 z            | 1.683 (0.211) | 5.177 (0.321) | 7.0 z            |
| 14–24.5                                    | 52.7 (1.14) | 74.4 (1.68) | 1.4 z            | 1.563 (0.126) | 4.729 (0.236) | 6.8 z            |
| 14–26                                      | 52.8 (0.67) | 72.8 (1.24) | 1.3 zyx          | 1.549 (0.088) | 4.358 (0.181) | 6.1 zyx          |
| 14–27.5                                    | 52.8 (0.48) | 64.1 (0.71) | 0.7 x            | 1.597 (0.068) | 2.908 (0.093) | 2.7 x            |

salmon (unadjusted survival of 95.9%), but 30 fish were lost due to thermal exposure (adjusted survival of 97.3%). Of the 30 fish lost due to thermal exposure, all but 2 (or 93.3%) were in the two systems that daily reached or exceeded 27°C ( $F_{14-27.5}$  and  $F_{12-27}$ ), and most of these fish died in the first 2 d of the experiment. For example, in  $F_{14-27.5}$ , where a total of 17 fish died from thermal exposure (adjusted survival rate of 83%), 13 died in the first 24 h and 3 died in the next 24 h, (i.e., 16 of the 17 fish [94%] died within 48 h of beginning the experiment). In  $F_{12-27}$ , where 11 fish died (adjusted survival rate of 88.9%), 5 died within 24 h, 5 died within 24–48 h, and 1 died with 48–72 h. No additional thermal mortalities were noted in either group for the remainder of the study. Further, there were only two other mortalities possibly related to thermal exposure in the fluctuating thermal regimes, one in  $F_{14-24.5}$  and one in  $F_{10-25}$ ; no mortalities occurred in the thermal regime that increased to 26°C. Tukey's HSD test of individual treatments showed that mortalities due to thermal exposures were significantly highest in  $F_{14-27.5}$ , followed by  $F_{12-27}$ ; both groups were significantly higher than all other groups ( $P < 0.05$ ).

In the fluctuating thermal regimes, the maximum daily growth rates was in  $F_{14-23}$  at 1.4% FL/d and 7.0% WT/d; second highest growth was in  $F_{14-24.5}$  at 1.4% FL/d and 6.8% WT/d (Table 4). The lowest growth occurred in the two fluctuating regimes that had daily temperature peaks of 27°C or higher:  $-0.7\%$  FL/d and  $2.7\%$  WT/d in  $F_{14-27.5}$  and  $0.8\%$  FL/d and  $3.4\%$  WT/d in  $F_{12-27}$ . These two groups were significantly lower than  $F_{14-23}$  and  $F_{14-24.5}$  (FL:  $P = 0.001$ – $0.043$ ; WT:  $P = 0.006$ – $0.023$ ).

Each fluctuating thermal regime was modeled separately, and we determined that a polynomial worked well as a function of the maximum temper-

ature (Figure 4). In fact, all three regime series did not have significantly different first-order and second-order parameters (FL:  $F_{2, 41} = 2.41$ ,  $P = 0.0678$ ; WT:  $F_{4, 35} = 1.98$ ,  $P = 0.1197$ ) and could be represented by a single model for fork length and a single model for weight. Further, the intercepts from the polynomial for each regime series (Table 5), fit with a simple linear model, had significant slopes associated with the acclimation temperature (FL:  $P = 0.009$ ; WT:  $P = 0.046$ ), and  $R^2$  values equal to 0.99. Thus, the best model for the daily growth rate in fork length was represented by a combination of the linear model based on the intercepts of the polynomial model and the remaining first and second order terms ( $R^2 = 0.90$ ), that is,

$$Y = -17.19 + 0.07 \cdot AT + 1.54 \cdot MT - 0.03 \cdot MT^2, \quad (3)$$

where AT is the acclimation temperature (°C) and MT is the maximum temperature (°C).

Similarly, the daily growth rate in weight was best represented by the model ( $R^2 = 0.857$ )

$$Y = -115.23 + 0.40 \cdot AT + 10.09 \cdot MT - 0.22 \cdot MT^2. \quad (4)$$

The growth rate in fork length and weight of fish in the fluctuating regimes were always significantly less ( $P < 0.001$  for both FL and WT) than their counterpart in the constant regime for the same (or similar) average temperature (Figure 5). In fact, the growth in the fork length and weight for the constant regime were 1.59 and 1.85 times, respectively, the fluctuating regime, even after removing the effect of average temperature and the variation in temperature (i.e., coefficient of variation) within each temperature regime.

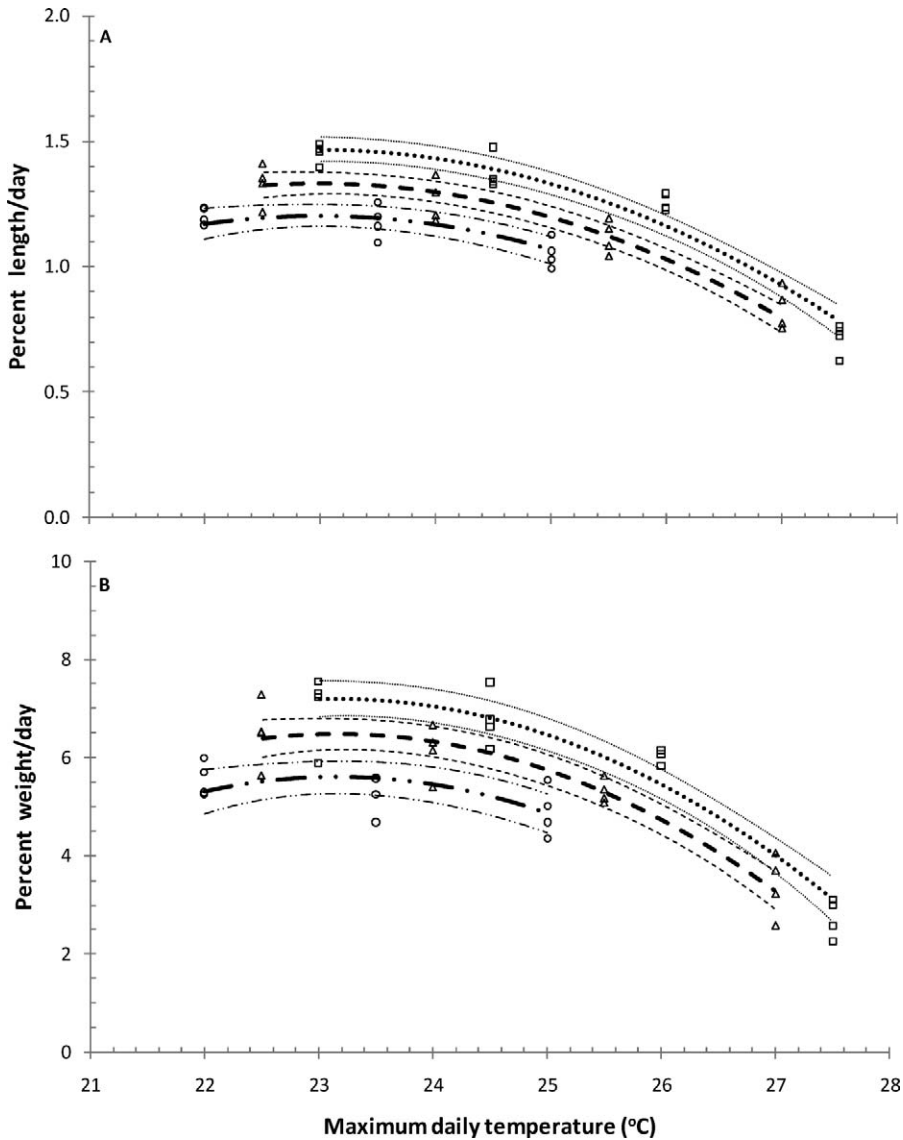


FIGURE 4.—Daily percent increases in (A) the fork length and (B) the weight of juvenile fall Chinook salmon as a function of the maximum daily temperature to which fish were exposed each day for 30 d. The different symbols represent the observed values at different acclimation temperatures (circles [10°C], triangles [12°C], and squares [14°C]); the thick lines depict the expected results based on regression analysis, the thin lines the 95% confidence intervals.

Discussion

In this study we tested the thermal tolerance of juvenile fall Chinook salmon using water temperature regimes in the laboratory that matched field conditions found in entrapment pools of the Snake River. This approach was different from ILT studies (Brett 1952; Elliot 1981) that expose test animals to rapid changes of constant test temperatures followed by extended holding (96 h or more). Our approach also differed

from CTM studies (Coutant 1973; Becker and Genoway 1979; Elliot and Elliot 1995) that expose fish to rapid temperature changes (18°C/h). Because neither ILT or CTM accurately reflects conditions fall Chinook salmon experience in entrapment pools of the Snake River, we used a modified CTM method to match the heating rate of the entrapment pools (1.5°C/h) and found that juvenile Snake River fall Chinook salmon acclimated to 10–14°C first lost equilibrium at mean

TABLE 5.—Regression results for fluctuating temperature regimes, based on the polynomial model of the daily percent change in fork length (FL) and weight of juvenile fall Chinook salmon as a function of the maximum temperature, with the first-order and second-order parameters shared.

| Parameter or statistic   | FL gain (%/d)  |                |                | Weight gain (%/d) |                 |                 |
|--------------------------|----------------|----------------|----------------|-------------------|-----------------|-----------------|
|                          | 10°C           | 12°C           | 14°C           | 10°C              | 12°C            | 14°C            |
| Best-fit values          |                |                |                |                   |                 |                 |
| Intercept ( $\beta_0$ )  | -16.5          | -16.4          | -16.3          | -111.3            | -110.3          | -109.7          |
| Slope1 ( $\beta_1$ )     | 1.5            |                |                | 10.1              |                 |                 |
| Slope2 ( $\beta_2$ )     | -0.03          |                |                | -0.22             |                 |                 |
| 95% confidence intervals |                |                |                |                   |                 |                 |
| Intercept ( $\beta_0$ )  | -21.5 to -11.5 | -21.4 to -11.4 | -21.3 to -11.2 | -148.0 to -74.7   | -147.0 to -73.7 | -146.3 to -73.0 |
| Slope1 ( $\beta_1$ )     | 1.1 to 2.0     |                |                | 7.1 to 13.1       |                 |                 |
| Slope2 ( $\beta_2$ )     | -0.04 to -0.03 |                |                | -0.3 to -0.2      |                 |                 |
| Goodness of fit          |                |                |                |                   |                 |                 |
| df                       | 39             |                |                | 39                |                 |                 |
| R <sup>2</sup>           | 0.900          |                |                | 0.857             |                 |                 |
| Mean square error        | 0.0007         |                |                | 0.005             |                 |                 |

temperatures of 26.8–27.3°C and half of the fish died at 27.4–27.9°C. These thermal tolerance values were higher than the published ILT of about 25°C for this species (Brett 1952; Orsi 1971). However, the CTM and ILT values for a species are not the same and comparisons between the two are confounded by the different exposure techniques. The upper temperature threshold associated with a loss of equilibrium or death that is determined using the CTM method (exposes fish to rising temperatures) will generally exceed the ILT value (extended exposure to an abrupt temperature change; Becker and Genoway 1979). The changing temperatures associated with CTM studies results in a lag between water temperature and core body temperature, so the water temperature passes the thermal tolerance boundary ahead of the core body temperature (Becker and Genoway 1979); this lag will increase with faster heating rates. Considering the thermal conditions that fall Chinook salmon are exposed to within Snake River entrapment pools, comparison of our results to the ILT value for Chinook salmon is not appropriate.

There have been surprisingly few CTM studies conducted on Chinook salmon (Myrick and Cech 2004). American River fall Chinook salmon acclimated to 19°C lost equilibrium at 28.8°C; the heating rate was 18°C/h (Cech and Myrick 1999). The CTM in the American River study may have been higher than our results because of differences between the two studies in acclimation temperature and heating rates (Kilgour and McCauley 1986). Higher acclimation temperatures and faster heating rates normally increase the thermal tolerance of salmonids. For example, studies on coho salmon *O. kisutch* parr by Becker and Genoway (1979) found that the CTM associated with a loss of equilibrium in two groups of fish heated at 18°C/h

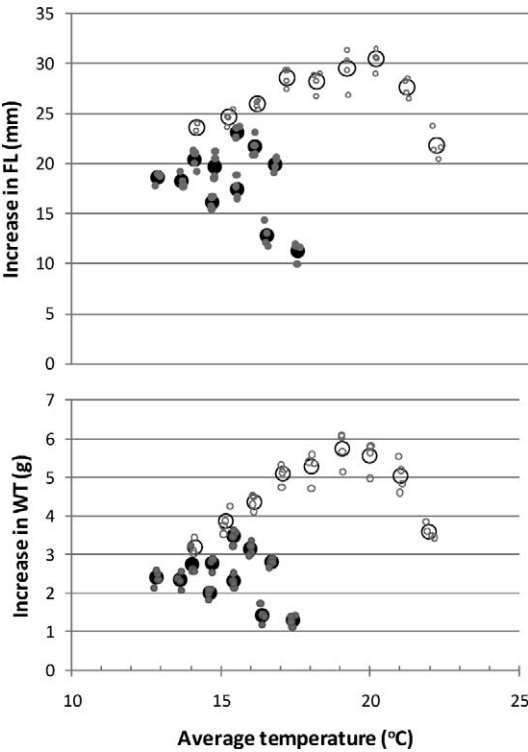


FIGURE 5.—Absolute increase in the fork length (FL) and weight (WT) of juvenile fall Chinook salmon as a function of the average daily temperature for 30 d. Fish were exposed to either a constant thermal regime (open circles) or a fluctuating thermal regime (filled circles); the small circles represent the responses in individual fish containers, the larger circles the averages for the exposure systems overall.

increased from 25.3°C to 29.7°C when acclimation temperatures were increased from 5°C to 15°C. In addition, when acclimation temperature was held constant at 15°C but heating rate increased from 1°C/h to 60°C/h, the CTM increased from 27.6°C to 29.6°C. As previously noted, animals that are heated too fast may experience a significant time lag between core body temperature and water temperature, allowing the animal to survive longer at elevated temperatures. However, at too slow heating rates, animals may be able to acclimate to temperatures during the test procedure, thereby increasing their thermal tolerance. The CTM associated with a loss of equilibrium and death in pumpkinseeds *Lepomis gibbosus* acclimated to 10°C was higher at a heating rate of 1°C/h than at 6°C/h; thermal tolerance then became higher as heating rate increased (Becker and Genoway 1979). It is likely that had we used a faster heating rate, the CTM of our fish would have increased above what we measured; this assumes that the 1.5°C/h rate that we used did not allow for acclimation to the test temperature.

We determined that exposure duration in both constant and fluctuating tests significantly affected mortality rates. Constant exposure for 30 d at 22°C resulted in little mortality, but up to 57% of fish died in the tests of thermal tolerance where temperatures were allowed to rise and hold constant at 25°C for up to 72 h. Snyder and Blahm (1970) showed that fingerling Chinook salmon acclimated to 10°C in the lower Columbia River could withstand a 4-h exposure to 21°C, but that a 37-min exposure to 26.7°C caused 100% mortality. Marine and Cech (2004) reared Sacramento River fall Chinook salmon at 21–24°C without mortality, but in an unpublished study (Rich 1987) cited by Myrick and Cech (2004), significant mortality occurred after 8 d at 24°C. In the fluctuating thermal regimes we tested for 30 d, short-term exposure (<3 h) of juvenile fall Chinook salmon to temperatures of 27°C or more resulted in mortality; this finding was in accordance with the results from our modified CTM trials. However, if the daily maximum temperature did not exceed 26°C and was followed by a rapid decline of 3°C/h, consistent with conditions in the entrapment pools, then the survival rates were nearly 100%; this was true regardless of the acclimation (base) temperature.

It is not the amplitude of the fluctuation that determines survival when temperatures are constant, but rather the extreme temperature that is reached (and likely the duration it is maintained). All Bonneville cutthroat trout *O. clarkii utah* survived after being exposed to a temperature fluctuation from 16°C to 26°C, even though the 7-d upper incipient lethal temperature was 24.2°C (Johnstone and Rahel 2003).

Dickerson and Vinyard (1999) observed high mortality of Lahontan cutthroat trout *O. clarkii henshawi* when exposed to a constant temperature of 26°C for 7 d, but no mortality over 7 d when temperatures fluctuated daily between 20°C and 26°C. Columbia River fall Chinook salmon were able to survive fluctuating temperatures from 12° to 26°C (heated at 2°C/h), but exposures to 26°C longer than a couple hours or exposure to temperatures above 26°C resulted in additional mortality (Mesa et al. 2002). Mesa et al. (2002) documented that fish exposed to fluctuating temperatures from 12°C to 26°C had a liver heat shock protein (HSP) 70 level that was 25 times that of control fish. It appears that HSPs provide an intracellular repair process that extends the time organisms can tolerate lethal temperatures, as long as the exposure is not too long or too high (Mesa et al. 2002; Schrank et al. 2003; Johnstone and Rahel 2003; Werner et al. 2005).

We found that Snake River fall Chinook salmon had a maximum growth rate of 11.2% WT/d at 20°C when fed to satiation. Although growth rates for Snake River fall Chinook salmon were much higher than other Chinook salmon, the temperature range for optimum growth was consistent with other studies. Fall Chinook salmon from the Sacramento River reared at 17–20°C had a maximum growth rate of 3.3% WT/d (Marine and Cech 2004). American River fall Chinook salmon grew at a maximum rate of 4.4% WT/d when fed to satiation at a constant 19°C (Myrick and Cech 2002). Brett et al. (1982) found that at satiated feeding levels, the optimum temperature for growth of juvenile Chinook salmon in the laboratory was around 19°C at which maximum growth rates were 3.2% WT/d. Cech and Myrick (1999) determined 19°C to be the temperature of optimum growth at rations associated with 100% satiation. Banks et al. (1971) determined 15.6°C to be the temperature of optimum growth in fingerling Chinook salmon fed to satiation, and although 18.3°C provided slightly higher growth in some tests prolonged exposure to 18.3°C was anticipated to be detrimental due to disease (Banks et al. 1971).

Fish in our study were reared on groundwater, which minimized their exposure to pathogens and parasites, causative agents of fish disease. Although we did not detect disease in our test fish, increased susceptibility of salmonids to disease organisms has been associated with elevated temperatures approaching 20°C (reviewed by Richter and Kolmes 2005). Juvenile Chinook salmon, coho, and steelhead *O. mykiss* infected with the bacteria *Aeromonas salmonicida* died at much higher rates when held at 20.5°C (93–100% mortality) than at 3.9°C (2% mortality) and 6.7°C (26% mortality); similar results were found with exposures to

*A. hydrophila* (Groberg et al. 1978). Chinook salmon, coho salmon, and steelhead did not die when exposed to *Flexibacter columnaris* if temperatures were less than 9.4°C, but suffered 70–100% mortality at 20.5°C (Holt et al. 1975). The optimum temperature for the infectious development of the microsporidian parasite *Enterocytozoon salmonis* in Chinook salmon occurred at 15°C and 18°C (Antonio and Hedrick 1995). However, juvenile Chinook salmon acclimated to 20°C were protected from lethal challenges using the hemoflagellate *Cryptobia salmositica* (Bower 1995) and juvenile coho salmon, sockeye salmon *O. nerka*, and steelhead infected with the causative agent of bacterial kidney disease were more susceptible at 6.7°C to 12.2°C than at 20.5°C (Sanders et al. 1978). Further, juveniles originating in waters where a disease is present may be more resistant than those from areas lacking that disease (Hemmingsen et al. 1986).

Our data and the studies cited above suggest that the optimal temperature for Chinook salmon growth is 16–20°C, provided food is not limited and other factors such as disease, predation, and competition have a minimal effect (Myrick and Cech 2004). Reduced food availability usually results in a downward shift of the optimum growth temperature in salmonids because more energy is needed to maintain higher metabolic rates that increase with a rise in temperature (Fry 1971). This “temperature-shunt” relationship (Brett et al. 1982) has been determined experimentally for many species, including sockeye salmon, brown trout *Salmo trutta*, and yellow perch *Perca flavescens* (Brett et al. 1969; Elliot 1976; Kitchell and Stewart 1977). Fall Chinook salmon in our study had access to unlimited food. Because the food supply in nature is not unlimited, the temperature for optimum growth of Snake River fall Chinook salmon in the wild is probably less than 20°C. For example, the optimum temperature for growth of the Chinook salmon studied by Brett et al. (1982) in the laboratory dropped from 19°C to 15°C in the wild, where fish were feeding at 60% satiation, a level more likely to occur in the Snake River. The authors predicted that river temperatures of 20°C would only increase body weight by 50% and that no growth would occur at 21.4°C.

In our study, temperatures above 20°C (the optimum for growth at unlimited rations) were detrimental to growth, whether in the constant or fluctuating thermal regimes. The percent growth in fork length and weight in the constant thermal regimes were nearly twice those in the fluctuating regimes, even though daily average temperatures were similar. Hokanson et al. (1977) found that diel temperature fluctuations were beneficial to rainbow trout *O. mykiss* growth, provided the fluctuations were no greater than the physiological

optimum (17.2°C); once fluctuations went above 17.2°C, growth and yield declined compared with the constant temperature regimes. These results suggest that despite an abundance of food, there is a point in the rising temperature regime below the upper thermal tolerance level where growth is stabilized or even reduced. In this case, energy input does not keep pace with needs for standard metabolism, digestion, and activity (Warren and Davis 1967). Further, even at satiated food levels, the feeding rate of salmonids declines as temperature increases. Donaldson and Foster (1941) found that sockeye salmon fingerlings ceased feeding when temperatures fluctuated from 17°C to 25.6°C. Baldwin (1957) determined that brook trout *Salvelinus fontinalis* had a reduced ability to capture prey when temperatures increased from 17°C to 21°C. In the fluctuating regimes we tested, fish were noticeably less active during feeding at times of the day when temperatures were above 20°C (1600–1700 hours) compared with times of day when temperatures were below 20°C.

We maintained oxygen at or near air saturation during the study, and total dissolved nitrogen was well below levels that would be harmful to juvenile salmonids. Thus, we have no way of knowing whether alteration of these variables may have affected temperature tolerance of Snake River fall Chinook salmon. Oxygen capacity of warm waters is somewhat lower than cooler waters, which may reduce the scope of activity for Chinook salmon (Warren and Davis 1967). Ebel et al. (1971) documented significantly reduced thermal tolerance of juvenile coho salmon, spring-run Chinook salmon, and steelhead when exposed to test water that was supersaturated (125–130% saturation) with nitrogen gas. Snyder and Blam (1970) determined that the upper incipient lethal temperature for fall Chinook salmon from the lower Columbia River was nearly 5°C below the 25.1°C value reported by Brett (1952), a finding the authors attributed to river water supersaturated with nitrogen gas (120% saturation). A warming pool of water may have a rising supersaturation level if the water is saturated with dissolved gases at the outset. However, the majority of the entrapment pools in the Snake River that are used by fall Chinook salmon are formed during load-following operations at river flows that do not result in spill. Consequently, dissolved gas levels are normally not elevated in the entrapment pools.

Indirect effects of warming temperature, including aberrant behavior or a loss of equilibrium, may make fish more susceptible to predation. Templeton and Coutant (1971) reviewed thermal studies conducted in the mid-Columbia River, Washington, and determined that predation increased on fall Chinook salmon when

they lost equilibrium due to temperature effects. Fall Chinook salmon reared at 21–24°C were more susceptible to predation by striped bass *Morone saxatilis* than at lower temperatures (Marine and Cech 2004). Juvenile Chinook salmon exposed to 26°C, the lowest test temperature used in his study, induced significantly different predation after exposure for 64 min but not after 16 or 32 min (Coutant 1973). However, following a thermal exposure consistent with entrapment pools on the Columbia River (acclimated to 12°C, heated at a rate of 2–4°C/h to 26°C, and returned to acclimation temperature within 1 h), Mesa et al. (2002) found juvenile fall Chinook salmon showed no increased vulnerability to predation by smallmouth bass *Micropterus dolomieu*.

Prolonged rearing at elevated temperatures has been shown to disrupt physiological processes associated with smoltification in salmonids (see reviews by McCullough et al. 2001; Myrick and Cech 2004; Richter and Kolmes 2005). However, the relationship between short-term exposure to elevated temperatures and inhibition of smoltification is not well understood. Further, environmental conditions and saltwater tolerance of subyearling fall Chinook salmon is complex because of a life history that places them in main-stem rivers at a time when temperatures are normally elevated (Zaugg 1981). In general, fall Chinook salmon can adapt to seawater challenges (one indicator of smoltification) in warmer temperatures than other salmonids. Subyearling fall Chinook salmon from the Sacramento River were able to adapt to seawater challenges at 14–16°C but sustained exposure to temperatures above 17°C inhibited gill Na<sup>+</sup>,K<sup>+</sup>-ATPase activity and reduced seawater adaptability (Marine and Cech 2004). Increasing acclimation temperature from 12°C to 15°C reduced salinity tolerance and gill Na<sup>+</sup>,K<sup>+</sup>-ATPase activity in coho salmon (Zaugg and McLain 1972; Adams et al. 1973, 1975; Wagner 1974). Smaller-sized salmonids may not be as adaptable to seawater tolerance (smoltification) as larger sizes, suggesting that temperatures inhibiting growth could also inhibit other physiological processes necessary for saltwater adaptation (reviewed by Myrick and Cech 2004). Little information is available to assess how smoltification is affected by short-term exposure to elevated temperatures like those Snake River fall Chinook salmon experience in the entrapment pools.

### Management Implications

Undisturbed natural habitats in the aquatic environment are relatively stable, moderate temperature changes being the normal condition. Exposure of organisms to constant and elevated temperatures consistent with the ILT method is a better indicator

of physiological impacts from elevated temperature than is the CTM method, where temperatures are constantly changing (Fry 1967; McCullough et al. 2001). Consequently, temperature standards within water quality criteria are often based on ILT rather than CTM methods (Becker and Genoway 1979). Researchers in the Columbia and Snake rivers have adopted this approach and often use 23°C as the upper lethal temperature level for fall Chinook salmon caught in entrapment pools along the Snake and Columbia rivers (Anglin et al. 2005; Brink and Chandler 2007). The 23°C threshold is based on an ILT of about 25°C (Brett 1952; Orsi 1971), adjusted downward with a 2°C safety factor (NAS 1972). However, artificially induced changes in flow that result in short-term and unstable changes in water temperature are increasingly common throughout our nation's waterways, including the Snake River below Hells Canyon Dam. In this case the ILT is not representative of thermal conditions fish experience in Snake River entrapment pools and studies using the CTM method are more suited to establishing acute thermal tolerance levels than are the classic ILT methods (Becker and Genoway 1979). Our results using a modified CTM method suggest that 26.8°C may be a more appropriate upper lethal temperature level when determining mortality in temporary entrapment pools of the Snake River; this temperature was the lowest maximum temperature that resulted in a loss of equilibrium in the thermal tolerance trials, where temperatures were progressively increased at 1.5°C/h.

At the onset of our study in 2008, an interim management agreement between National Oceanic and Atmospheric Administration Fisheries and Idaho Power Company required increasing flow releases from Hells Canyon Dam to reconnect the entrapment pools to the river channel when the average daily temperature in the entrapment pools exceeded 16°C for 3 d or if the instantaneous temperature exceeded 18°C for 4 h (Brink and Chandler 2007). The management guidelines for daily average temperature not to exceed 16°C was probably meant to keep long-term average temperatures in a thermal zone where overall fitness is maximized. This logic is reasonable, but our results suggest that maximum temperatures, rather than average temperatures, are better indicators of growth and survival. For example, mortality was significantly higher and growth was significantly lower in the group exposed daily to temperatures that fluctuated from 12°C to 27°C ( $F_{12-27}$ ), even though daily average temperatures were around 16°C (Table 1). Further, juvenile fall Chinook salmon with access to unlimited food grew best at 20°C, well above the 16°C guideline

(however, if food was limited, this threshold would be lower).

Although not specifically stated, the use of a maximum temperature is probably meant to provide protection to fall Chinook salmon from peak temperatures that approach lethal limits. The results from our study suggest 18°C is conservatively low. For example, exposure to constant temperatures up to 22°C for 30 d did not result in mortality, and survival to sustained temperatures of 25°C was variable, fish dying 1.7–22.5 h after temperatures reached 25°C. Our data along with other studies of fall Chinook salmon (Cech and Myrick 1999; Mesa et al. 2002) suggest 26°C may be the upper level at which fall Chinook salmon can survive, provided temperatures do not stay at this level for more than 1 h.

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